

Notes 5

Commands for navigating the file system

LS

Definiton

- is is used for listing files and directions.
- By default it will list the current directory when no directory is specified.
- Listing means to see what is inside a directory.
- I use this command whenever I want to "open and see" what is inside a given folder in my Linux system.
- I can also use this command when I need to see some information (metadata) of a file. For example, when it was last edited, its size, permissions, etc.

FORMULA/USAGE

- `ls + option + directory(ies) or file to list`

EXAMPLES

- See all the options of the ls command:
 - `ls --help`
- List all the files including hidden files in current directory:
 - `ls -A`
- Long list a directory
 - `ls -lA ~/Pictures`
- Long list multiple directories excluding group and owner information, with human readable file size and sorted in reverse order.
 - `ls -lhG Documents/ Pictures/`

```

ma@cis106:~$ ls --options
ls: unrecognized option '--options'
Try 'ls --help' for more information.
ma@cis106:~$ ls --help
Usage: ls [OPTION]... [FILE]...
List information about the FILES (the current directory by default).
Sort entries alphabetically if none of -cftuvSUX nor --sort is specified.

Mandatory arguments to long options are mandatory for short options too.
  -a, --all                do not ignore entries starting with .
  -A, --almost-all        do not list implied . and ..
  --author                 with -l, print the author of each file
  -b, --escape              print C-style escapes for nongraphic characters
  --block-size=SIZE        with -l, scale sizes by SIZE when printing them;
                           e.g., '--block-size=M'; see SIZE format below
  -B, --ignore-backups     do not list implied entries ending with ~
  -c                        with -lt: sort by, and show, ctime (time of last
                           change of file status information);
                           with -l: show ctime and sort by name;
                           otherwise: sort by ctime, newest first
  -C                        list entries by columns
  --color[=WHEN]           color the output WHEN; more info below
  -d, --directory           list directories themselves, not their contents
  -D, --dired               generate output designed for Emacs' dired mode
  -f                        same as -a
  --classify[=WHEN]        append indicator (one of */>@|) to entries WHEN
                           likewise, except do not append '*'
  --file-type              across, horizontal (-x), commas (-m), long (-l),
                           single-column (-1), verbose (-l), vertical (-C)
  --full-time              like -l --time-style=full-iso
                           like -l, but do not list owner
  -g                        group directories before files
  --group-directories-first
                           in a long listing, don't print group names
  -G, --no-group           with -l and -s, print sizes like 1K 234M 2G etc.
  -h, --human-readable      with -l and -s, print sizes like 1K 234M 2G etc.

```

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ma@cis106:~$ ls -lA ~/Pictures/
total 4
drwxrwxr-x 2 ma ma 4096 Feb 20 15:58 wallpaper
ma@cis106:~$ ls -lhG Documents/ Pictures/
ls: cannot access 'Pictures/': No such file or directory
ma@cis106:~$

```

PWD

Definition

- It is a command in Linux/Unix used to display the absolute path of the current directory you are in.
- The “working directory” is simply the folder/directory in which your terminal session is currently operating.
- The absolute path is the full path from the root directory / to your current folder.
- Essentially, `pwd` tells you: “Where am I in the filesystem right now?”

FORMULA/USAGE

- The general usage is very simple:
 - `pwd [options]`

EXAMPLES

- Suppose your directory structure is:
 - `/home/user/Documents/Projects`
- You navigate to the Projects folder:
 - `cd /home/user/Documents/Projects`
- Now you want to know where you are:
 - `pwd`
- Output:
 - `/home/user/Documents/Projects`
- Using the options with symbolic links:
 - `pwd -L # Shows the logical path`
 - `pwd -P # Shows the physical path`

DEFINITION

- It is a command in Linux/Unix used to navigate between directories in the filesystem.
- The “working directory” is the folder your terminal is currently operating in, and `cd` lets you move from one folder to another.

FORMULA/USAGE

- The general syntax is:
 - `cd [directory_path]`

EXAMPLES

- Suppose your directory structure is:
 - `/home/user/Documents/Projects`
 - `/home/user/Documents/Notes`
- You are in `/home/user/Documents`:
 - `pwd`
 - `# Output: /home/user/Documents`
- Go to `Projects` folder (relative path):

- `cd Projects`
- `pwd`
- `# Output: /home/user/Documents/Projects`
- Go up one directory (parent folder):
 - `cd ..`
 - `pwd`
 - `# Output: /home/user/Documents`
- Go to `Notes` folder using absolute path:
 - `cd /home/user/Documents/Notes`
 - `pwd`
 - `# Output: /home/user/Documents/Notes`
- Go back to the previous directory:
 - `cd -`
 - `pwd`
 - `# Output: /home/user/Documents/Projects`

What is a variable?

A name that stores a value which can be used later in a program or terminal session.

How do I use a variable?

Assign a value to a name and reference it using `$` in the terminal or script.

What is an environment variable?

A special variable that affects the behavior of processes and is available system-wide.

What is a user defined variable?

A variable created by the user in a shell session or script, usually only available in that session.

What is the root directory?

The top-most directory in the Linux filesystem, represented by `/`.

What does “Parent Directory” mean?

The directory located one level above the current directory.

What does “Current working directory” mean?

The directory you are currently in within the terminal session.

What is an absolute path? Include an example

The full path from the root directory `/` to a file or folder, independent of your current location.

What is a relative path? Include an example

A path to a file or folder that is relative to the current working directory.

What is the difference between “Your home directory” and “The home directory”?

- Your home directory: The specific directory assigned to your user account.
- The home directory: The generic concept of a home directory that contains all users' directories.